

LOGIC AND PROOFS

Predicates and Quantifiers

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01 September, 2025

OUTLINE

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Propositional logic cannot adequately express the meaning of all statements in mathematics and in natural language.

Predicate - a proposition that depends on the value of some variables.

For example,

- “ x is prime” will be true or false depending on the value of x .
- “ x is greater than 3”
- “ $x = y + 3$ ”
- “Computer x is under attack by an intruder”.

The Propositional Function

- “ x is greater than 3”, has 2 parts, (1) the variable x of the statement, and (2) the **predicate** “is greater than 3”, a property of the statement.
- We can denote the statement “ x is greater than 3” by $P(x)$, where P denotes the predicate “is greater than 3” and x is the variable.
- $P(x)$ is the value of the **propositional function** P at x . When a value is assigned x , $P(x)$ becomes a proposition with a truth value.

Examples:

1. Let $P(x)$ be the statement $x > 3$. What are the truth values for $P(4)$ and $P(2)$?
2. Let $Q(x, y)$ denote: student x has missed test y for CSI131.
 - Assume only Maria missed a test for CSI131 and she missed Test 2.
 - What are the truth values of: $Q(\text{Maria}, 1)$, $Q(\text{John}, 2)$, $Q(\text{Maria}, 2)$?
3. Let $A(y)$ denote the statement: "External disk y is infected by a virus". Suppose that there are only two external disks, Sahara and Kingston and that none of the disks is infected by a virus. What are the truth values of $A(\text{Kingston})$ and $A(\text{Sahara})$?
 - $A(\text{Kingston})$ is the statement: External disk Kingston is infected by a virus, which is false.
 - $A(\text{Sahara})$ is the statement: External disk Sahara is infected by a virus which is also false.

Predicate Calculus

- the area of logic that deals with predicates and quantifiers. Two ways to form propositions from predicates:

- Assigning values to parameters: - seen already.
- **by quantification:** Universal and Existential quantification.

Quantifiers

Now we are going to turn a propositional function into a proposition using a **quantifier**.

Two most widely-used quantifiers:

- “FOR ALL” $\forall$ UNIVERSAL QUANTIFIER.
- “THERE EXISTS” $\exists$ EXISTENTIAL QUANTIFIER.

e.g. $\forall x P(x)$ means for all x , $P(x)$ is true.

e.g. $\exists x P(x)$ means there exists some x , such that $P(x)$ is true.

The Quantifiers

	Universal $\forall$	Existential $\exists$
	"For ALL"	"THERE EXISTS"
When True?	When $P(x)$ is true for every x in the domain.	There is an x in the domain for which $P(x)$ is true.
When False?	There is an x in the domain for which $P(x)$ is false.	When $P(x)$ is false for every x in the domain.
	$\forall x P(x) \equiv P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$	$\exists x P(x) \equiv P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)$

The Universal Quantifier $\forall$

The statement $\forall x P(x)$ tells us that the proposition $P(x)$ must be true for all values of x in the **Domain of Discourse/Universe**.

Let $P(x)$ represent " $x > 0$ ". Find each truth value for $\forall x P(x)$:

- U is $\mathbb{Z}$.
- U is $\mathbb{Z}^+$.

The Existential Quantifier $\exists$

The statement $\exists x P(x)$ tells us that the proposition $P(x)$ is true for some values of x in the Domain of Discourse/Universe.

Let $P(x)$ represent " $x > 0$ ". Find each truth value for $\exists x P(x)$:

- U is $\mathbb{Z}$.
- U is $\mathbb{Z}^-$.

Examples:

1. Let $P(x)$ be " $x^2 > 0$ " if the domain consists of all integers. Find the truth values of:
 - $\forall x P(x)$ $0 \in \mathbb{Z}$, and $0^2 > 0$ is FALSE. Therefore, $\forall x P(x)$ is False.
 - $\exists x P(x)$ $1 \in \mathbb{Z}$, and $1^2 > 0$ is TRUE. Therefore, $\exists x P(x)$ is True.
2. Let $P(x)$ be " $x^2 < 0$ " if the domain consists of all integers. Find the truth values of:
 - $\forall x P(x)$ FALSE. $(-2)^2 < 0$. Therefore, $\forall x P(x)$ is False.
 - $\exists x P(x)$ FALSE. Therefore, $\forall x P(x)$ is False.

Practice:

1. Let $P(x)$ be " $x + 1 = 2x$ " if the domain consists of all integers. Find the truth values of:
 - $\forall x P(x)$ **FALSE**. Counterexample $4 + 1 \neq 2 \cdot 4$. Therefore, $\forall x P(x)$ is False.
 - $\exists x P(x)$ **True**.
2. Let $P(x)$ be " $x^2 < 16$ " if the domain consists of all integers. Find the truth values of:
 - $\forall x P(x)$ **FALSE**. Counterexample $(5)^2 \not< 0$. Therefore, $\forall x P(x)$ is False.
 - $\exists x P(x)$ **TRUE** for $-4 < x < 4$. Therefore, $\forall x P(x)$ is True.

The Uniqueness Quantifier

The statement, $\exists! x P(x)$ tells us that the proposition $P(x)$ is true for exactly one value of x in the domain of discourse/universe.

Give the truth value for $\exists! x P(x)$ for each proposition in the domain of all integers.

- $P(x)$ represents " $2x = 4$ " **TRUE**
- $P(x)$ represents " $2x > 4$ " **FALSE**
- $P(x)$ represents " $2x = 3$ " **FALSE, $1.5 \notin \mathbb{Z}$**

Precedence of Quantifiers

The quantifiers $\forall$ and $\exists$ have higher precedence than all logical operators from propositional calculus.

e.g. for $\forall x P(x) \vee A(y)$ we find the truth value of $\forall x P(x)$, then find the truth value of $A(y)$ and finally apply the $\vee$ to the two results.

Quantifiers with Restricted Domains

This allows for values of a parameter to be restricted. The condition is specified immediately after the parameter. e.g.

1. $\forall x < 0 (x^2 > 0)$: Note that the value of x is restricted to < 0 , This statement says: “for every real number x that is less than 0, $x^2 > 0$. This is equivalent to: $\forall x (x < 0 \rightarrow x^2 > 0)$. NB this is the same as the universal quantification of an implication.
2. $\exists y > 0 (y^2 = 9)$: note that y is also restricted. The statement means: “there exists a real number y that is greater than 0 such that $y^2 = 9$.” This is equivalent to: $\exists y (y > 0 \wedge y^2 = 9)$. Note that this is equivalent to the existential quantification of a conjunction

Binding Variables

When a quantifier is used on the variable x , we say that this occurrence of the variable is **bound**, otherwise it is **free**.

In $\exists x(x + y = 1)$, x is bound by the existential quantification $\exists x$, but y is free.

NEGATING AND TRANSLATING WITH QUANTIFIERS

Let $P(x)$ be the statement: “ x has taken a course in programming” for the domain of students in your class.

Negating Quantifiers		
Quantifier	Statement	Equivalent to
$\neg\forall xP(x)$	There is a student in my class who has not taken a course in programming.	$\exists x\neg P(x)$
$\neg\exists xP(x)$	All students in my class have not taken a course in programming.	$\forall x\neg P(x)$

De Morgan's Laws for Quantifiers

- $\neg \exists x P(x) \equiv \forall x \neg P(x)$
 - True when $P(x)$ is False for every x
 - False when there is an x for which $P(x)$ is True.
- $\neg \forall x P(x) \equiv \exists x \neg P(x)$
 - True when there is an x for which $P(x)$ is False.
 - False when $P(x)$ is True for every x .

Negate the Statements

1. There is an honest politician.

- $H(x)$ represents “ x is honest” for the domain of politicians.

$$\exists x H(x)$$

$$\neg \exists x H(x) \equiv \forall x \neg H(x)$$

- Every politician is dishonest, or All politicians are not honest.

2. All Africans eat chicken.

- $C(x)$ represents “ x eats chicken” for the domain of all Africans.

$$\forall x C(x)$$

$$\neg \forall x C(x) \equiv \exists x \neg C(x)$$

- Not every African eats chicken.

Translate statements: Domain

Some student in this class has visited Namibia.

1. Domain: Students in class.

- $M(x)$ represents “ x has visited Namibia”.
- $\exists x M(x)$

2. All people.

- $M(x)$ represents “ x has visited Namibia”
 - $C(x)$ represents “ x is a student in this class”
- $\exists x (M(x) \wedge C(x))$

Nested Quantifiers and Negations

Definition:

Nested quantifier Two quantifiers are nested if one is within the scope of the other. e.g.

$\forall x \exists y (x + y = 0)$ “Every real number has an additive inverse.”

$\forall x \exists y (x + y = 0)$ is the same as $\forall x Q(x)$, where $Q(x)$ is $\exists y P(x, y)$, where $P(x, y)$ is $x + y = 0$.

Let $P(x, y)$ denote “ $xy = yx$ ”. Assume the domain is real numbers.

1. Is $\forall x \forall y P(x, y)$ true? **TRUE** “For all real number x , and for all real numbers y , $xy = yx$.”
2. Is $\forall y \forall x P(x, y)$ true? **TRUE**

Quantifications of Two Variables.		
<i>Statement</i>	<i>When True?</i>	<i>When False?</i>
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y .

Practice

Let $Q(x)$ denote " $x + y = 5$ ". Assume the domain is real numbers.

1. Is $\forall x \exists y Q(x, y)$ true? "For all real number x , there exists a real number y , such that $x + y = 5$. TRUE
2. Is $\exists y \forall x Q(x, y)$ true? "There exists a real number y such that for all real number x , $x + y = 5$. FALSE

Practice

Let $\mathbf{U}$ be real numbers, and $P(x, y)$ denote " $x \cdot y = 0$ ". Find the truth values for the following:

1. Is $\forall x \forall y P(x, y)$ true? **FALSE**
2. Is $\forall x \exists y P(x, y)$ true? **TRUE**
3. Is $\exists x \forall y P(x, y)$ true? **TRUE**
4. Is $\exists x \exists y P(x, y)$ true? **TRUE**

Practice

Let $\mathbf{U}$ be real numbers, and $P(x, y)$ denote " $x/y = 1$ ". Find the truth values for the following:

1. Is $\forall x \forall y P(x, y)$ true? **FALSE**
2. Is $\forall x \exists y P(x, y)$ true? **FALSE** "False because if x is 0, it doesn't work."
3. Is $\exists x \forall y P(x, y)$ true? **FALSE**
4. Is $\exists x \exists y P(x, y)$ true? **TRUE**

Translating with Nested Quantifiers

Translate “The sum of two positive integers is always positive” into a logical expression.

We can write the above as

“For all positive integers x and y , $x + y > 0$ ”.

1. $\forall x \forall y ((x + y) > 0)$, domain $\mathbb{Z}^+$.
2. $\forall x \forall y ((x > 0) \wedge (y > 0)) \rightarrow (x + y > 0)$, domain $\mathbb{Z}$.
3. Let $P(x)$ denote $(x + y > 0)$, we can replace $(x + y > 0)$ with $P(x)$ in 1. and/or 2. above.

Translating with Nested Quantifiers

Translate into statements involving nested quantifiers “Every real number except zero has a multiplicative inverse” into a logical expression. A **multiplicative inverse** of a real number x is a real number y such that $xy = 1$.

We can rewrite the above as

“For every real number x except 0, x has a multiplicative inverse”, or better “For every real number x , if $x \neq 0$, then there exists a real number y such that $xy = 1$, i.e.,

$$\forall x(x \neq 0) \rightarrow \exists y(xy = 1).$$

Translating from Nested Quantifiers into English

Translate the statement

$\forall x(C(x) \vee \exists y(C(y) \wedge F(x, y)))$ into English

where $C(x)$ is “ x has a computer, ” $F(x, y)$ is “ x and y are friends”, and the domain for both x and y consists of all students in your school.”

Solution

The statement says that for every student x in your school, x has a computer or there is a student y such that y has a computer and x and y are friends. In other words, every student in your school has a computer or has a friend who has a computer.

Translating English Sentences into Logical Expressions

Express the statement “If a person is female and is a parent, then this person is someone’s mother” as a logical expression involving predicates, quantifiers with a domain consisting of all people, and logical connectives.

Solution

The statement can be expressed as “For every person x , if person x is female and person x is a parent, then there exists a person y such that person x is the mother of person y .” We introduce the propositional functions $F(x)$ to represent “ x is female,” $P(x)$ to represent “ x is a parent,” and $M(x, y)$ to represent “ x is the mother of y .” The original statement can be represented as:

$$\forall x((F(x) \wedge P(x)) \rightarrow \exists y M(x, y))$$

which is the same as

$$\forall x \exists y((F(x) \wedge P(x)) \rightarrow M(x, y))$$

The Negating Nested Quantifiers

Let $P(x, y)$ denote “ $x = -y$ ”. Find the negation of $\forall x \exists y P(x, y)$

No negation to precede a quantifier

$$\begin{aligned}\neg(\forall x \exists y P(x, y)) &\equiv \exists x \neg(\exists y P(x, y)) \\ &\equiv \exists x \forall y \neg P(x, y).\end{aligned}$$

Therefore, the negation is “There exists a real number x , such that for all real numbers y , $x \neq -y$.”

Express the Negation of the Statement

Solution

By successively applying De Morgan's laws for quantifiers, we can move the negation in $\neg\forall x\exists y(xy = 1)$ inside the qualifiers.

$$\begin{aligned}\neg(\forall x\exists y(xy = 1)) &\equiv \exists x\neg(\exists y(xy = 1)) \\ &\equiv \exists x\forall y\neg(xy = 1).\end{aligned}$$

Because $\neg(xy = 1)$ can be expressed more simply as $xy \neq 1$, we conclude that our negated statement can be expressed as $\exists x\forall y(xy \neq 1)$."

Logic Programming

- **Prolog - Programming with Logic:** is a programming language designed to reason using rules of predicate logic.
- It has two types of statements: **Prolog Facts** and **Prolog Rules**.
- Prolog facts define predicates.
- Prolog rules define new predicates using those already defined using Prolog facts.

Prolog program given facts about:

- instructors of each course. This is represented by the predicate: *instructor(p,c)* - to represent that professor *p* is the instructor for course *c*.
- which classes students are enrolled. This is represented by the predicate: *enrolled(s,c)* - to represent that student *s* is enrolled in course *c*.

Example:

- Study the Prolog Facts in a program given below
 - *instructor(chan, math273)*
 - *instructor(patel, ee222)*
 - *instructor(groosman, cs301)*
 - *enrolled(kevin, math273)*
 - *enrolled(juan, ee222)*
 - *enrolled(juan, cs301)*
 - *enrolled(kiko, math273)*
 - *enrolled(kiko, cs301)*
- Use lowercase for entries as Prolog considers names beginning with uppercase to be variables.

- We can define a new predicate, $teaches(p,s)$ to represent that professor p teaches student s .
- This can be defined using the Prolog rule:
 - $teaches(P, S) :- instructor(P,C), enrolled(S, C)$
- This means: $teaches(p,s)$ is true if there is a class c such that professor p is instructor of class c and student s is enrolled in class c .
- The comma represent $\wedge$ and for $\vee$ its ; in Prolog.

- Prolog answers queries using rules and facts given.
 - The query: *?enrolled(kevin, math273)* evaluates to **yes** due to the fact given earlier.
 - The query: *?enrolled(X, math273)* produces **kevin, kiko**. Prolog determines all possible values of X for which *enrolled(X, math273)* has been included as a Prolog fact.
 - The query: *?teaches(X,juan)* establishes professors who are instructors to courses being taken by juan. It will produce: **patel, grossman**.

The End